

CALCULUS 1

August 2017

Time : 2 hours

Candidates may attempt ALL questions in Section A and at most TWO questions in Section B. Each question should start on a fresh page.

SECTION A (40 marks)

Candidates may attempt ALL questions being careful to number them A1 to A4.

A1. (a) Solve the following inequality $4|x| < 7x - 6$.

(b) Show that $|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.

[4,4]

A2. (a) Define what is meant by saying that l is the **limit** of a sequence $\{u_n\}$. [2]

(b) Evaluate each of the following limits.

(i) $\lim_{n \rightarrow \infty} \left(\frac{\sqrt{3n^2 - 5n + 4}}{2n - 20} \right),$ (ii) $\lim_{n \rightarrow \infty} (n - \sqrt{n^2 + 1}).$ [3,3]

A3. (a) Estimate the value of $\sqrt[5]{36}$ using differentials. [4]

(b) Use the Mean Value Theorem to prove that $\ln(1 + x) < x$ for $x > 0$. [5]

A4. (a) Show that $\int x^m \ln x \, dx = x^{m+1} \left(\frac{\ln x}{m+1} - \frac{1}{(m+1)^2} \right), m \neq -1.$ [6]

(b) Find each of the following integrals.

(i) $\int \cos^5 x \, dx$ (ii) $\int \frac{x}{x+2} \, dx.$ [5,4]

SECTION B (60 marks)

Candidates may attempt TWO questions being careful to number them B5 to B8.

- B5.** (a) (i) Prove that if $0 < a < b$, then $a^2 < b^2$. [4]
 (ii) If $a^2 < b^2$, does it necessarily imply that $a < b$? If so, prove it and if not give an example that illustrates your answer. [4]
 (b) (i) Define in terms of ε and δ , what is meant by saying that a function $f(x)$ is continuous at a point $x = x_0$. [2]
 (ii) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} x^2 + 1, & x \geq 2 \\ ax + b, & -1 < x < 2 \\ 5 - 2x, & x \leq -1, \end{cases}$$

where a and b are constants. If f is continuous on $\mathbb{R}$, find the values of a and b . [6]

- (c) (i) State the Mean Value theorem. [4]
 (ii) Use the Mean Value Theorem or otherwise to prove that

$$1 - \frac{a}{b} < \ln \left(\frac{b}{a} \right) < \frac{b}{a} - 1$$

if $0 < a < b$. [6]

- (iii) Hence or otherwise show that

$$\frac{1}{6} < \ln(1.2) < \frac{1}{5}.$$

[4]

- B6.** (a) (i) Define what is meant by $\lim_{n \rightarrow \infty} u_n = \infty$.
 (ii) Show using the definition, that $\lim_{n \rightarrow \infty} (3n + 7) = \infty$.
 (iii) Prove that $\lim_{n \rightarrow \infty} \frac{c}{n^p} = 0$, where $c \neq 0, p > 0$. [3,4,4]
 (b) (i) What is the relationship between differentiability and continuity?
 (ii) Hence prove this relationship. [3,6]
 (c) (i) If $e^{xy} + y \ln x = \sin 2x$, find $\frac{dy}{dx}$.
 (ii) If $y = x^x, x > 0$, find $\frac{dy}{dx}$. [6,4]

B7. (a) Evaluate each of the following limits.

$$(i) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} \quad (ii) \lim_{x \rightarrow 0} x^{\sin x}. \quad [4,6]$$

(b) (i) Prove that if $f(x)$ is continuous on $[a, b]$, differentiable on (a, b) and if $f'(x) \geq 0$ for every x in (a, b) , then $f(x)$ is monotonic increasing on $[a, b]$.

(ii) Verify Rolle's Theorem for $f(x) = x^2(1 - x)^2$, $0 \leq x \leq 1$. [4,6]

(c) Show that the function $f(x) = \frac{ax + b}{cx + d}$ is one-to-one or injective if $ad - bc \neq 0$. Find its inverse stating the domains of both the function and its inverse. What happens if $ad - bc = 0$? [10]

B8. (a) Evaluate each of the following integrals.

$$(i) \int \frac{\sqrt{x}}{\sqrt{x} + 1} dx \quad (ii) \int \sin^{-1} x dx. \quad [5,5]$$

(b) Find the area of the region bounded by the graph of $f(x) = 1 - x^2$, $x = 0$, $x = 1$ and the x -axis by calculating the limit of the Riemann Sums. [10]

(c) Find

$$\int \frac{dx}{(x^2 + 1)(x + 1)}. \quad [10]$$

END OF QUESTION PAPER